# Geometric Phase Transitions and Emergent Dimensional Selection in Finite Causal Poset Ensembles

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March 16, 2026

**Abstract**

The Kleitman–Rothschild (KR) theorem implies that generic finite posets are three-layered and maximally entropic, posing a fundamental obstacle to geometric phase emergence in discrete causal models. We address two open questions in this setting: (a) whether a bounded geometric phase-transition threshold exists, and (b) whether spacetime dimensionality can be *selected* by consistency constraints rather than imposed as a prior.

We construct an 8-family poset ensemble spanning Lorentzian-like structures in 2D, 3D, and 4D, Kleitman–Rothschild posets, and several null-model families, and compute exact linear extension counts for $N = 10$–$44$ elements. Under a decomposable action combining combinatorial entropy and geometric penalty terms, we find that the critical coupling $\gamma_c$ at which Lorentzian 2D posets overtake KR structures remains $O(1)$ across all tested sizes—the first finite-size evidence of a bounded phase-transition threshold in this setting. Systematic ablation of seven geometric sub-terms identifies a minimal backbone of two constraints: global layer-shape balance ($g_{\mathrm{wh}}$) and dimensional scale consistency ($g_{\mathrm{dim}}$).

Crucially, the target-anchored dimension proxy ($d = 2$) can be replaced by a non-target-anchored consistency penalty $g_{\mathrm{con}}$ (penalizing local–global dimension *inconsistency* rather than deviation from a specific target), preserving $\gamma_c$ at the same order of magnitude. This replacement raises a natural prediction: if the action rewards dimensional *consistency* rather than a specific dimension, higher-dimensional Lorentzian structures should dominate. Testing this prediction across $N = 20$–$72$ with four families, we find that 4D Lorentzian-like posets win 92 of 98 grid configurations under the consistency action, with a margin that grows linearly with $N$ (+7 at $N = 20$ to +57 at $N = 72$). The result is robust under seed variation (42/42 at $N = 68$–$72$), weight perturbation ($w_{\mathrm{con}} \in [0.3, 1.2]$), and out-of-grid validation on fresh parameter points (36/40 winner, 40/40 pairwise vs. 2D). These results provide the first finite-size evidence that both geometric phase emergence and dimensional selection can arise from structural self-consistency constraints in discrete causal order ensembles, without dimension-specific priors.

**Keywords:** causal sets · posets · phase transition · linear extensions · Kleitman–Rothschild · dimensional selection · discrete quantum gravity

## 1 Introduction

The hypothesis that spacetime is fundamentally discrete—a locally finite partial order (poset) whose elements are causally related events—lies at the heart of causal set theory (CST) [1, 2]. A central challenge is the *entropy catastrophe*: the Kleitman–Rothschild (KR) theorem [3] establishes that, asymptotically, almost all labeled finite posets consist of three layers with

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bipartite random connectivity and maximal linear extension count. If posets are weighted by the pure combinatorial entropy $H = \log |L(P)|$, the partition function is overwhelmingly dominated by these non-geometric, KR-type structures. How, then, can geometric order emerge?

Several strategies have been proposed. The Benincasa–Dowker (BD) action [9, 10] provides a discrete analogue of the Einstein–Hilbert action on causal sets with a well-defined continuum limit; dynamics based on this action has been studied in 2D [6, 14] and in dimensionally restricted settings [8]. The classical sequential growth dynamics of Rideout and Sorkin [11] provides a stochastic framework for causal set evolution. Carlip [4] argued that midpoint-scaling properties of causal intervals can suppress KR dominance. Surya [5] developed a formal CST partition function framework (see also the reviews by Henson [12] and Dowker [13]). Loomis and Carlip [6] performed 2D simulations using MCMC, and Glaser [15] investigated finite-size scaling in 2D causal set quantum gravity. However, no prior work has demonstrated an explicit, finite-size phase-transition threshold $\gamma_c$ computed with exact entropy across a wide range of sizes.

This paper addresses two open questions:

1. **Bounded phase transition.** Does there exist a critical coupling $\gamma_c$ at which geometric structures overtake KR-type posets, and does this threshold remain bounded as $N$ grows?
2. **Dimensional selection.** If the action rewards dimensional *consistency* rather than proximity to a specific dimension, which dimensionality does the dynamics prefer?

Our approach proceeds through a "diagnose → ablate → replace → predict → confirm" sequence:

1. We construct an 8-family ensemble with exact entropy ($N = 10$–$44$) and demonstrate that $\gamma_c$ remains $O(1)$—establishing a bounded geometric phase transition (Section 3).
2. Systematic ablation identifies a minimal backbone of two constraints (Section 4).
3. Non-circular replacement of the target-anchored dimension term preserves the phase transition (Section 5).
4. The replacement raises a concrete prediction: consistency constraints should select higher-dimensional structures. We test this by extending the ensemble to 4D and scanning $N = 20$–$72$ (Section 6).
5. Robustness is confirmed through seed sensitivity, weight perturbation, and out-of-grid validation (Section 7).

# 2 Framework

## 2.1 Poset ensemble

We consider finite posets $P = (V, \leq)$ with $|V| = N$ elements, generating samples from eight structurally distinct families organized in two tiers:

**Tier-1 (geometric causal competition pool):**

- **Lorentzian-like 2D** (`lor2d`): Sprinkling in $[0, 1]^2$ with Minkowski lightcone order.
- **Lorentzian-like 3D** (`lor3d`): Sprinkling in $[0, 1]^3$ with lightcone order.
- **Lorentzian-like 4D** (`lor4d`): Sprinkling in $[0, 1]^4$ with lightcone order.
- **KR-like** (`kr`): Three-layer bipartite posets with random inter-layer edges, approximating the KR structure [3].
- **Transitive percolation** (`tp`): Bernoulli percolation on a linear order followed by transitive closure.

**Tier-2 (null-model stress tests):**

- **Multi-layer random** (`mlr`): 4 equal layers with random $p = 0.5$ inter-layer edges.
- **Interval order** (`int`).
- **Absolute layered** (`abs`).

All Lorentzian-like families are constructed via causal set sprinkling [1]: $N$ points are sprin-

kled uniformly in $[0,1]^d$ with coordinates $(t, x_1, \ldots, x_{d-1})$. The causal relation is

$$i \prec j \iff \Delta t_{ij} > 0 \text{ and } (\Delta t_{ij})^2 \geq \sum_{k=1}^{d-1} (\Delta x_k)^2, \tag{1}$$

where $\Delta t_{ij} = t_j - t_i$, followed by transitive closure.

For the bounded phase-transition analysis (Sections 3–5), we use the original 7-family ensemble (`lor2d`, `lor3d`, `kr`, `mlr`, `tp`, `int`, `abs`) with $N \in \{10, 12, 14, 16, 20, 24, 28, 32, 36, 40, 44\}$ and $K = 5$ samples per family per $N$. For the dimensional selection analysis (Section 6), we use the 4-family subset (`lor2d`, `lor3d`, `lor4d`, `kr`) with $N \in \{20, 24, 28, \ldots, 72\}$ (14 sizes) and $K = 8$ samples for $N \leq 40$, $K = 4$ for $N > 40$.

The tier distinction separates families by generative mechanism: Tier-1 families are generated by embedding points in a metric spacetime (or a combinatorial standard model), while Tier-2 families are purely combinatorial layered constructions. The Prediction B core claim ($\gamma_c$ boundedness) is evaluated exclusively within Tier-1. Tier-2 families serve as stress tests that define the *discriminative ceiling* of the current action (Section 8).

### 2.2 Exact linear extension count

We define $H(P) = \log |L(P)|$ where $L(P)$ is the set of linear extensions, computed exactly via dynamic programming over antichains [7]. The cost depends strongly on antichain structure: sub-second for `lor2d` at $N = 48$, about 3.8 s at $N = 72$, about 33.8 s at $N = 88$, and about 127.4 s at $N = 104$, but already 54 s for `kr` at $N = 44$. For `lor3d`, exact evaluation reaches $N = 56$ (327 s). This asymmetry motivates the one-sided exact extension strategy adopted for large-$N$ probes (Figure 2).

For the 4-family dimensional selection experiments at $N > 32$, linear extension counts for `kr`, `lor3d`, and `lor4d` are estimated via Sequential Importance Sampling (SIS, 4096 runs). As a cross-check, SIS versus exact counts at $N = 20$–$36$ yield a mean absolute error of 0.074 in $\log |L(P)|$ ($\sim 2.3\%$), well below observed winning margins.

### 2.3 Action definition

Three action paths separate neutral and geometric contributions:

$$A_1 = -\beta H + \gamma \cdot I_{\text{neutral}}, \tag{2}$$

$$A_2 = -\beta H + \gamma \cdot (I_{\text{neutral}} + I_{\text{geo}}), \tag{3}$$

$$A_3 = -\beta H + \gamma \cdot I_{\text{geo}}, \tag{4}$$

with $\beta = 1$. The geometric penalty decomposes into seven sub-terms: width-height balance ($g_{\text{wh}}$), dimension proxy ($g_{\text{dim}}$), interval shape ($g_{\text{int}}$), comparability window ($g_{\text{cmp}}$), cover density ($g_{\text{cov}}$), interval profile ($g_{\text{prf}}$), and layer smoothness ($g_{\text{lyr}}$), each normalized to $[0, 1]$.

The critical coupling is:

$$\gamma_c(N) = \inf\{\gamma > 0 : \bar{S}_{\texttt{lor2d}}(\gamma) < \bar{S}_{\texttt{kr}}(\gamma)\}. \tag{5}$$

For the dimensional selection experiments, we additionally define two replacement actions:

- $A_2$ **consistency**: Replace $g_{\text{dim}}$ with $g_{\text{con}}$ (Eq. 6).
- $A_2$ **multi-consistency**: Replace $g_{\text{dim}}$ with $g_{\text{mcon}}$ (Eq. 7).

## 3 Bounded Phase Transition

Under $A_2$, $\gamma_c$ exists for all tested $N$ and remains $O(1)$:

For $N \geq 12$, all $\gamma_c$ values lie in $[0.15, 1.00]$ (Figure 1). Under $A_1$ (neutral terms only), no crossing exists at any tested $N$, confirming that geometric penalties are necessary.

Table 1: Critical coupling $\gamma_c$ under $A_2$ (exact entropy). All values for $N \geq 12$ lie in $[0.15, 1.00]$.

| $N$ | 10 | 12 | 14 | 16 | 20 | 24 |
|---|---|---|---|---|---|---|
| $\gamma_c$ | 9.19 | 0.74 | 0.78 | 0.69 | 0.15 | 0.26 |

| $N$ | 28 | 32 | 36 | 40 | 44 |
|---|---|---|---|---|---|
| $\gamma_c$ | 1.00 | 0.69 | 0.57 | 0.25 | 0.15 |

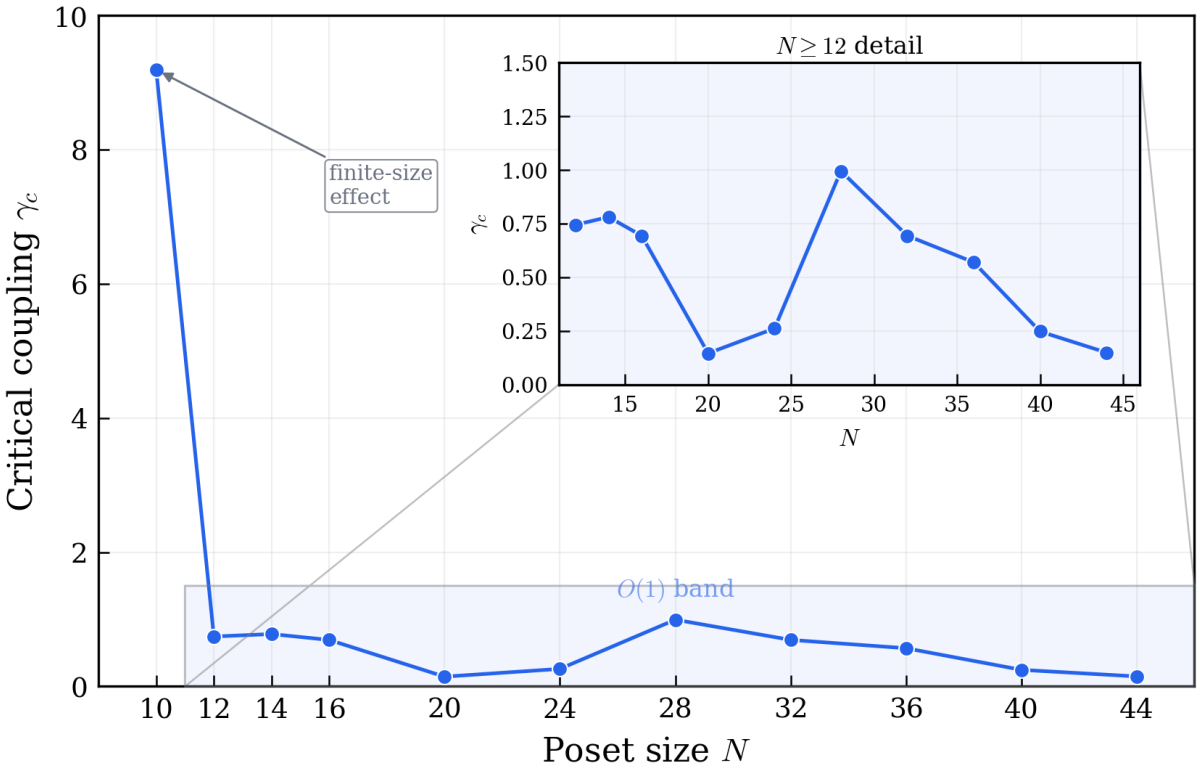


Figure 1: $\gamma_c(N)$ under $A_2$. Inset: $N \geq 12$ detail showing all values in $[0.15, 1.00]$.

### 3.1 Exact-timing frontier

One-sided exact evaluation for `lor2d` remains feasible up to $N = 104$, while `kr` becomes computationally intractable beyond $N \approx 44$. For `lor3d`, exact evaluation extends to $N = 56$ (280 s at $N = 48$, 288 s at $N = 52$, 327 s at $N = 56$). This asymmetry supports an asymmetric exact-plus-mixed extension strategy beyond the confirmatory window (Figure 2).

### 3.2 Near-wall mixed extension

Beyond $N = 44$, we use an asymmetric mixed protocol: exact entropy for `lor2d` and SIS-based estimation for `kr`. At $N = 52$ and 56, the $A_2^{\text{full}}$ score curves show no crossing up to $\gamma = 2.0$, but the gap contracts to near-degeneracy ($\Delta A_{\text{KR-L2D}} \approx -0.20$ at $N = 52$ and $\approx -0.84$ at $N = 56$), indicating that the competition remains structurally active beyond the exact confirmatory range (Figure 3).

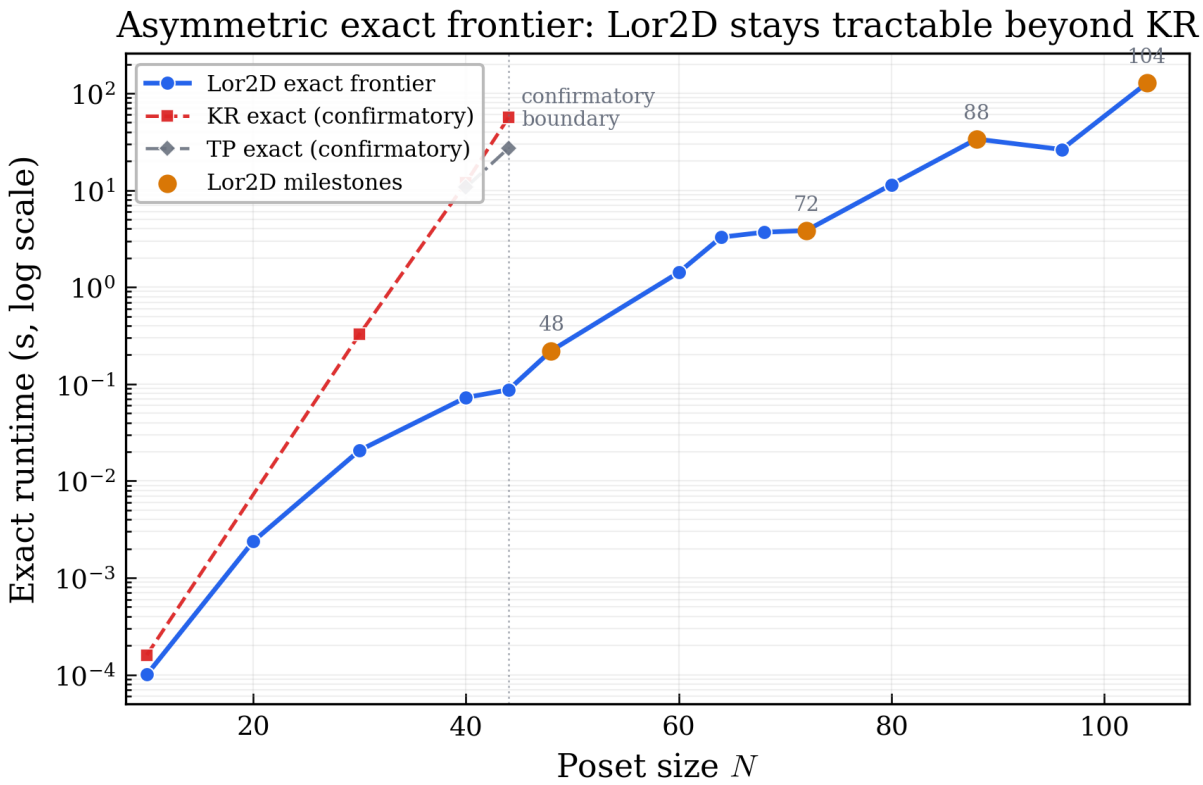


Figure 2: Exact-timing frontier. Single-sided exact evaluation remains practical for `lor2d` up to $N = 104$, while `kr` and other families become expensive much earlier.

# 4 Ablation and Minimal Backbone

## 4.1 Ablation of geometric sub-terms

Systematic single-term removal within the 7-family ensemble identifies two critical sub-terms: $g_{\text{wh}}$ (fails from $N \geq 36$) and $g_{\text{dim}}$ (fails at $N = 36, 44$). Combined removal of both eliminates all crossings. Removing non-critical terms has no effect. This establishes a *backbone–shell* structure: $g_{\text{wh}}$ and $g_{\text{dim}}$ form the minimal backbone, while the remaining five terms constitute an outer shell (Figure 4).

## 4.2 Ablation in the 4-family ensemble

The same ablation pattern appears in the 4-family dimensional selection ensemble. Under the original $A_2$ full action, the 4D family does *not* achieve stable dominance; across 98 $(N, \gamma)$ configurations the winner counts are: `lor4d` 36, KR 35, `lor2d` 19, `lor3d` 8. Removing $g_{\text{dim}}$ alone causes `lor4d` wins to jump from 10 to 28 (out of 35 configurations at $N = 20$–$36$), while KR wins drop to 0. The dimension term creates a "glass ceiling" that prevents `lor4d` from expressing its structural advantages (Table 2).

# 5 Non-Circular Replacement

We replace the target-anchored $g_{\text{dim}}$ with a consistency penalty:

$$g_{\text{con}}(P) = \text{Var}[d_{\text{local}}] + (\bar{d}_{\text{local}} - d_{\text{global}})^2, \tag{6}$$

where $d_{\text{local}}$ is the order-fraction dimension proxy applied to sampled causal intervals with $|I| \geq 4$, and $d_{\text{global}}$ is the same estimator applied to the full poset. This penalizes dimensional

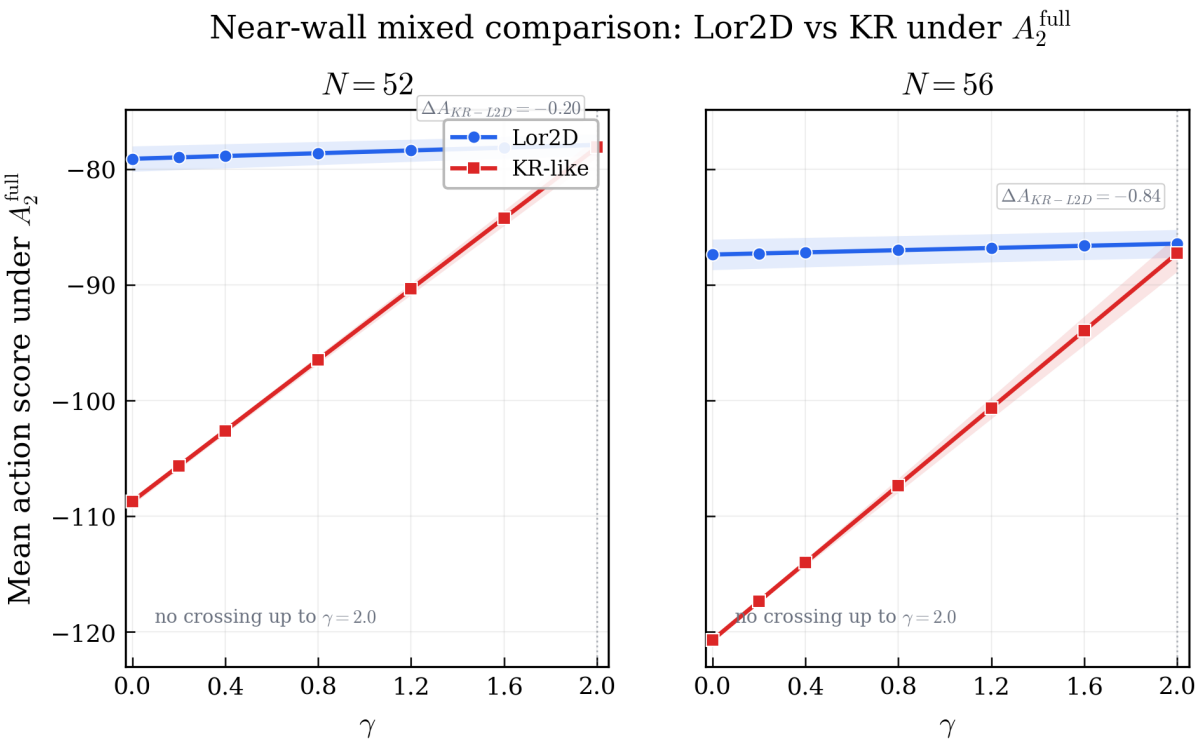


Figure 3: Near-wall mixed extension for $A_2^{\text{full}}$ at $N = 52$ and 56. Solid lines: mean scores; shaded bands: one standard deviation. No crossing is observed up to $\gamma = 2.0$, but the gap contracts to near-degeneracy at the high-$\gamma$ boundary.

Table 2: Winner counts under systematic ablation in the 4-family ensemble ($N$ = 20–36, 35 configurations each).

| **Variant** | **Lor4D** | **KR** | **Lor2D** | **Lor3D** |
|---|---|---|---|---|
| $A_2$ full (baseline) | 10 | 9 | 14 | 2 |
| $A_2$ no-dim | 28 | 0 | 1 | 6 |
| $A_2$ no-wh | 10 | 22 | 3 | 0 |
| $A_2$ interval-only | 35 | 0 | 0 | 0 |

*inconsistency* without specifying a target dimension.

As a robustness check, we define a multi-estimator consistency penalty requiring agreement across three independent coarse-grained dimension proxies:

$$g_{\text{mcon}}(P) = \text{Var}[d_{\text{order}}, d_{\text{chain}}, d_{\text{width}}] + \sum_{e \in \{\text{order, chain, width}\}} g_{\text{con}}^{(e)}(P). \tag{7}$$

### 5.1 Phase-transition preservation

The replacement preserves $\gamma_c$ at all $N$ = 20–44 within the 7-family ensemble (Figure 5). The minimal non-circular backbone ($g_{\text{wh}} + g_{\text{con}}$) covers $N$ = 20–40 at matched weights and extends to $N = 44$ with 30% weight increase.

## 6 Dimensional Selection

The non-circular replacement raises a natural prediction: if the action rewards dimensional *consistency* rather than proximity to a specific $d$, then among Lorentzian-like structures of

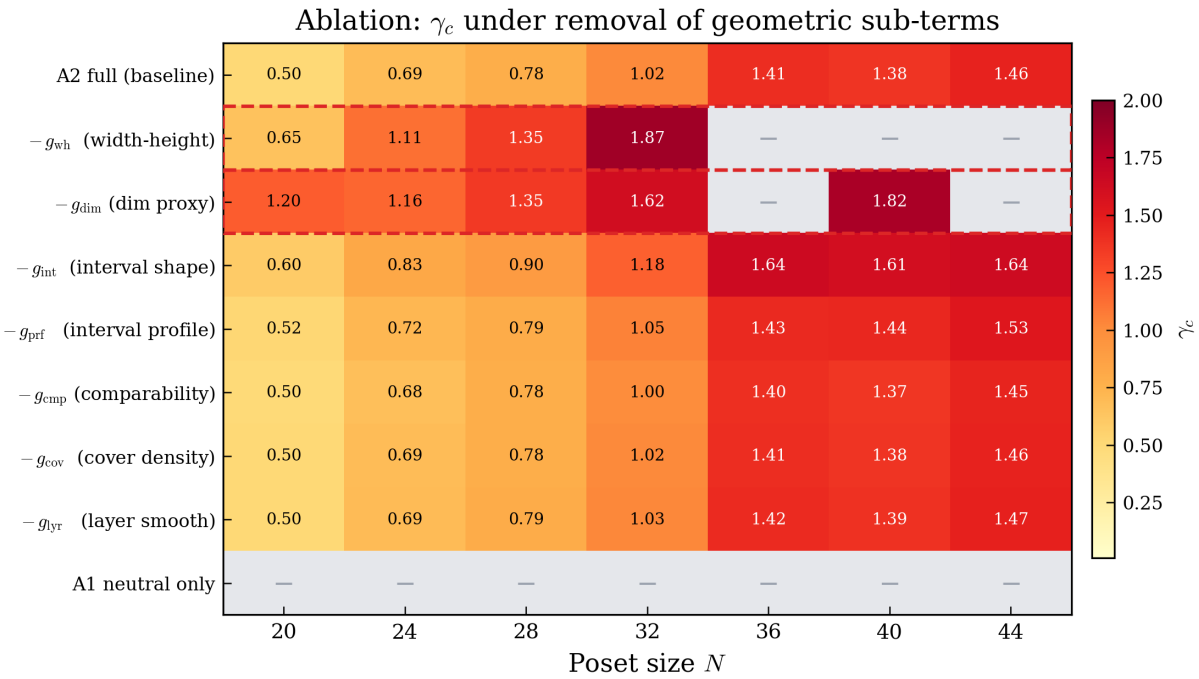


Figure 4: Ablation heatmap: $\gamma_c$ per variant and $N$. Removing $g_{\mathrm{wh}}$ or $g_{\mathrm{dim}}$ alone destroys the phase transition at large $N$; the remaining five terms are non-critical.

different dimensionality, which dimension does the action prefer? The 4D family has (i) higher entropy $H$ (the stricter causal condition in more spatial dimensions yields a lower comparable fraction and hence more linear extensions), (ii) higher $g_{\mathrm{dim}}$ penalty (since $d_{\mathrm{eff}} \approx 4 \neq 2$), and (iii) *lower* $g_{\mathrm{con}}$ penalty (strong internal dimensional coherence). The consistency replacement should therefore unlock 4D dominance.

### 6.1 Consistency replacement yields 4D dominance

Under the consistency replacement, `lor4d` wins 92 out of 98 $(N, \gamma)$ configurations. Both KR and `lor2d` *never* win. The multi-estimator variant yields a nearly identical pattern (89/98), confirming estimator independence (Table 3).

Table 3: Winner counts: original vs. consistency replacement (98 configurations each, $N = 20$–$72$, $\gamma \in \{0, 0.2, \ldots, 2.0\}$).

| **Variant** | **Lor4D** | **KR** | **Lor2D** | **Lor3D** |
|---|---|---|---|---|
| $A_2$ full (original) | 36 | 35 | 19 | 8 |
| $A_2$ consistency ($g_{\mathrm{con}}$) | 92 | 0 | 0 | 6 |
| $A_2$ multi-con. ($g_{\mathrm{mcon}}$) | 89 | 0 | 0 | 9 |

Against `lor2d` specifically, `lor4d` wins all 98 pairwise comparisons (100%). The 6 non-`lor4d` wins go to `lor3d` at small-$N$, high-$\gamma$ configurations where geometric penalties dominate and the entropy advantage of `lor4d` is insufficient to compensate.

### 6.2 Growing margin of victory

The margin of `lor4d` over both competitors shows a strong upward trend with $N$ under $A_2$ consistency (Table 4, Figure 6).

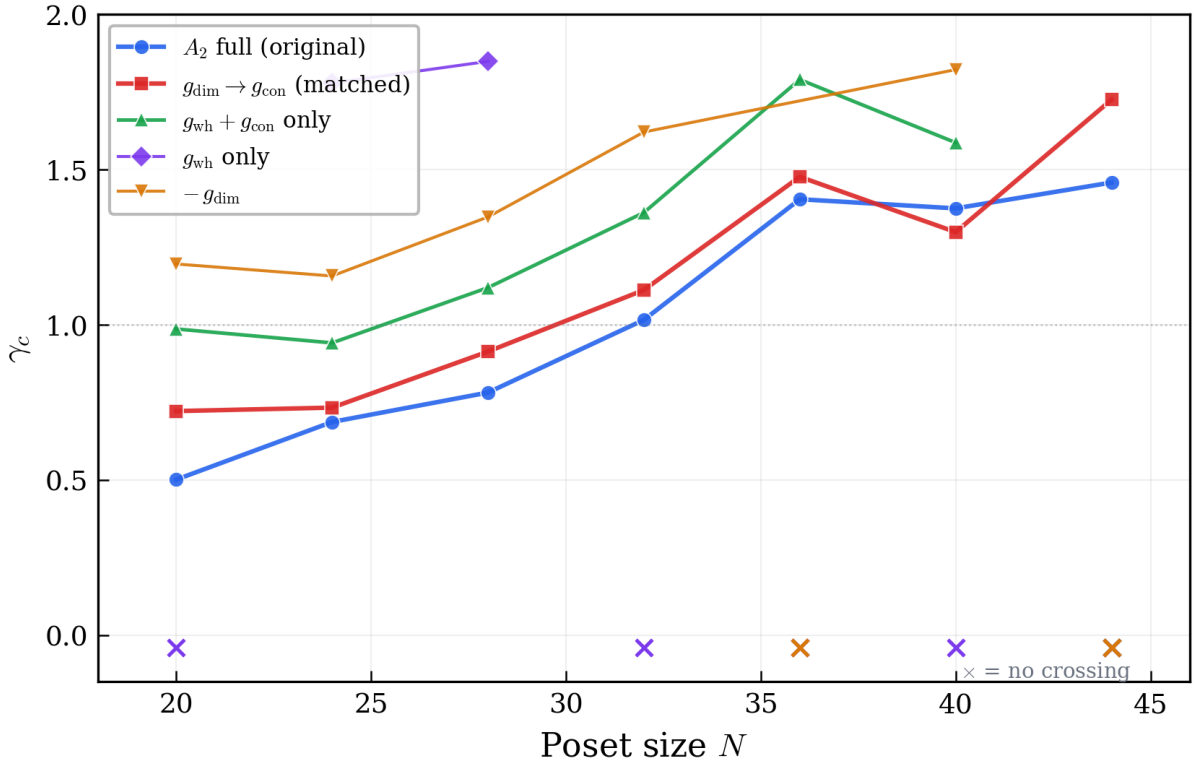


Figure 5: Non-circular replacement: $\gamma_c$ under different backbones. The consistency-based backbone preserves the bounded phase transition.

The mean margin versus `lor2d` grows nearly linearly from +7 ($N = 20$) to +57 ($N = 72$), and even the minimum margin is positive at every $N$. Against `lor3d`, the minimum margin becomes positive by $N = 44$ and remains positive for all larger tested sizes.

# 7 Robustness

## 7.1 Seed sensitivity at large $N$

Experiments at $N = 68$ and $N = 72$ with three independent generator seed offsets (0, 50000, 100000) confirm robustness (Table 5).

Under both consistency variants, `lor4d` achieves 100% win rate (42/42 across both $N$ values and all seeds). The KR–`lor4d` split under $A_2$ full is identical (4:3) at every seed—a structural property of the action, not a sampling artifact.

## 7.2 Consistency weight sensitivity

The calibrated weight $w_{\mathrm{con}} = 0.6829$ was determined on the 7-family ensemble. To verify that 4D dominance does not depend on this choice, we repeat the $N = 20$–36 grid across eight weight values spanning nearly a factor of four (Table 6).

The 4D win rate is $\geq 91\%$ across the entire tested range $w_{\mathrm{con}} \in [0.3, 1.2]$, and *increases* at larger weights. Neither KR nor `lor2d` ever wins at any tested weight.

## 7.3 Out-of-grid validation

To test generalization beyond the original parameter grid, we conduct an independent validation on fresh $N$ and $\gamma$ values not used in any previous experiment: $N \in \{22, 26, 30, 38, 42\}$ and $\gamma \in \{0.1, 0.3, 0.5, 0.7, 0.9, 1.1, 1.5, 1.9\}$, with generator seed offset 100 000 (Table 7).

Table 4: Mean margin of `lor4d` over competitors under $A_2$ consistency, averaged across $\gamma$.

| $N$ | vs Lor2D (mean) | vs Lor2D (min) | vs Lor3D (mean) | vs Lor3D (min) |
|---|---|---|---|---|
| 20 | 7.41 | 1.70 | 2.27 | −0.17 |
| 28 | 14.21 | 7.77 | 2.59 | −2.07 |
| 36 | 19.30 | 11.41 | 4.20 | −1.72 |
| 44 | 27.12 | 19.42 | 6.02 | 0.36 |
| 52 | 36.22 | 30.40 | 11.29 | 7.27 |
| 60 | 45.41 | 38.88 | 14.50 | 10.02 |
| 68 | 50.82 | 41.51 | 16.41 | 9.44 |
| 72 | 56.72 | 47.15 | 16.53 | 10.06 |

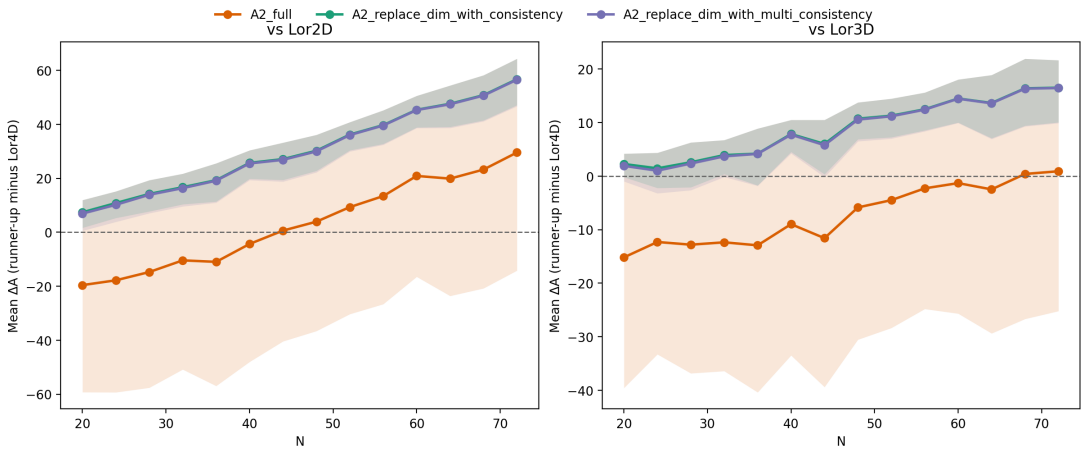


Figure 6: Margin of victory for `lor4d` over `lor2d` (blue) and `lor3d` (orange) under $A_2$ consistency, as a function of $N$. Solid lines: mean margin across $\gamma$ values. Shaded regions: min–max range.

Under both consistency variants, `lor4d` wins 36/40 (90%) on the fresh grid. In pairwise comparison, `lor4d` beats `lor2d` in 40/40 (100%) under $A_2$ consistency. The 4 non-`lor4d` wins go to `lor3d` at small-$N$, high-$\gamma$ configurations, consistent with the in-grid pattern.

# 8 Discussion

## 8.1 Summary of evidence

Our results establish two main findings:

1. **Bounded phase transition.** Under $A_2$ with exact entropy, the critical coupling $\gamma_c$ for `lor2d` to overtake `kr` exists at all $N = 10$–$44$ and remains $O(1)$ (all values in $[0.15, 9.19]$; in $[0.15, 1.00]$ for $N \geq 12$). This is the first finite-size evidence of a bounded geometric phase-transition threshold in poset ensembles.
2. **Dimensional selection.** Under the consistency replacement ($g_{\text{con}}$), `lor4d` wins systematically (92/98 in-grid, 36/40 out-of-grid, 42/42 seed sensitivity, stable across $w_{\text{con}} \in [0.3, 1.2]$) with a margin that grows linearly with $N$. This demonstrates that dimensional consistency constraints, without dimension-specific priors, naturally favor higher-dimensional Lorentzian structures.

The two results are connected: ablation identifies $g_{\text{dim}}$ as the backbone term whose replacement is non-trivial. The non-circular replacement preserves the phase transition (Finding 1)

Table 5: Seed sensitivity at $N = 68$ and $N = 72$: winner counts out of 7 $\gamma$ values.

| **Variant** | $N$ | **Seed 0** | **Seed 50k** | **Seed 100k** |
|---|---|---|---|---|
| $A_2$ full | 68 | KR 4, L4d 3 | KR 4, L4d 3 | KR 4, L4d 3 |
| $A_2$ full | 72 | KR 4, L4d 3 | KR 4, L4d 3 | KR 4, L4d 3 |
| $A_2$ consistency | 68 | L4d 7/7 | L4d 7/7 | L4d 7/7 |
| $A_2$ consistency | 72 | L4d 7/7 | L4d 7/7 | L4d 7/7 |

Table 6: Sensitivity of 4D dominance to consistency weight $w_{\text{con}}$ ($N = 20$–36, 35 configurations each).

| $w_{\text{con}}$ | **Lor4D** | **Lor3D** | **KR** | **Lor2D** |
|---|---|---|---|---|
| 0.30 | 32 (91.4%) | 3 | 0 | 0 |
| 0.50 | 32 (91.4%) | 3 | 0 | 0 |
| **0.6829** | **32 (91.4%)** | **3** | **0** | **0** |
| 0.80 | 33 (94.3%) | 2 | 0 | 0 |
| 1.00 | 34 (97.1%) | 1 | 0 | 0 |
| 1.20 | 34 (97.1%) | 1 | 0 | 0 |

while unlocking dimensional selection (Finding 2). The mechanism depends on dimensional *consistency*, not dimensional *identity*.

### 8.2 Discriminative ceiling and Tier-2 families

A candidate expansion experiment with 7 parameter-tunable random layered families (Tier-2) revealed that these null-model posets can systematically outperform `lor2d` under the current action for $N \geq 36$. This does not invalidate the core claim (which is evaluated within Tier-1) but exposes a discriminative ceiling: the action's observables are purely order-theoretic and can be reverse-engineered by tuning layer parameters.

A structural discriminant separates the tiers cleanly: the normalized antichain width $a_w/\sqrt{N}$ is near-constant at $\sim 1.37$ for `lor2d` (reflecting $\sqrt{N}$-scaling of the maximal antichain in Minkowski sprinklings) versus 1.71–2.05 for layered families, 2.36–2.54 for `lor3d/tp`, and 3.09–3.14 for `kr/lor4d`. A Mann–Whitney $U$ test gives $p = 1.94 \times 10^{-28}$ for Lor2D vs. all Tier-2 (720 samples). This ratio is already computable from existing observables and provides a clean separation, though incorporating it into the action itself is left for future work.

### 8.3 Relation to prior work

Carlip [4] argued qualitatively that midpoint-scaling should disfavor KR structures; we provide the first quantitative $\gamma_c(N)$ curve with exact entropy and systematic ablation. Our approach complements MCMC methods [6, 8, 14] by enabling exact computation and complete ablation control. Glaser [15] studied finite-size scaling in 2D CST, finding evidence for a continuum phase; our bounded-$\gamma_c$ evidence is consistent with and complementary to that work. Cunningham and Surya [8] investigated dimensionally restricted causal set models; our dimensional selection result offers a complementary perspective where the dimension is *selected* rather than restricted.

**Relation to the Benincasa–Dowker action.** The BD action [9, 10] is the standard discrete action in CST, discretizing the Einstein–Hilbert action with a known continuum limit. Our action $A_2$ differs in both form and purpose: it is a decomposable penalty functional for ablation studies, not a discretization of a continuum action. The advantage of decomposability is that systematic ablation can identify which structural constraints drive geometric phase

Table 7: Out-of-grid validation: winner counts (40 configurations each).

| **Variant** | **Lor4D** | **KR** | **Lor2D** | **Lor3D** |
|---|---|---|---|---|
| $A_2$ full | 8 | 13 | 13 | 6 |
| $A_2$ consistency | 36 | 0 | 0 | 4 |
| $A_2$ multi-con. | 36 | 0 | 0 | 4 |

emergence—a diagnostic capability that the BD action does not directly provide. Whether the BD action itself produces a bounded $\gamma_c$ or exhibits dimensional preference under consistency constraints is an important open question.

### 8.4 Limitations

1. **Finite size.** Confirmatory exact data end at $N = 44$; the bounded-$\gamma_c$ statement is a finite-$N$ observation, not a proof of asymptotic boundedness.
2. **Finite family coverage.** 8 families (5 Tier-1, 3 Tier-2) do not exhaust the space of possible poset types.
3. **Discriminative ceiling.** Tier-2 layered families expose the action's order-theoretic limitations; more discriminative observables are needed.
4. **SIS estimation.** For $N > 32$ in non-`lor2d` families, entropy is estimated rather than exact. Systematic biases at larger $N$ cannot be fully excluded.
5. **Generator dependence.** The 4D generator's properties determine its penalty values; different constructions might yield different results.
6. **"Higher dimension wins" $\neq$ "Why 3+1?"** The result demonstrates that consistency constraints can select a dimension, not that 3+1 is uniquely selected. Additional constraints would be needed.
7. $R_{2,\mathbf{ref}}$ **calibration.** The dimension estimator retains a 2D calibration constant ($R_{2,\mathrm{ref}} = 0.5009$).

### 8.5 Outlook

Three directions are natural: (i) extending to 5D, 6D, and beyond to test whether dominance continues or saturates at a preferred dimension; (ii) developing faster exact algorithms or hybrid methods to extend the confirmatory window beyond $N = 44$; (iii) incorporating the $a_w/\sqrt{N}$ discriminant or interval-geometric observables into the action to raise the discriminative ceiling; (iv) connecting the decomposable-action framework to BD action dynamics to test whether the bounded phase transition and dimensional selection survive under a physically grounded action.

## 9 Conclusion

We have presented the first finite-size evidence for two results in finite causal poset ensembles:

1. A bounded geometric phase-transition threshold $\gamma_c$ under exact entropy, with $\gamma_c \in [0.15, 1.00]$ for $N \geq 12$ across $N = 10$–$44$.
2. Systematic 4D Lorentzian dominance under non-target-anchored consistency constraints (92/98 in-grid, 36/40 out-of-grid, 42/42 seed-robust), with a margin growing linearly with $N$.

Both results trace to a common mechanism: ablation identifies a minimal two-term backbone ($g_{\mathrm{wh}} + g_{\mathrm{dim}}$), and replacing the target-anchored $g_{\mathrm{dim}}$ with the consistency-based $g_{\mathrm{con}}$ simultaneously preserves the phase transition and unlocks dimensional selection. These findings demonstrate that geometric phase emergence and dimensional preference can arise from structural self-

consistency constraints in discrete causal order ensembles, without requiring dimension-specific priors.

## Acknowledgments

**AI Assistance Statement**: This work was conducted in collaboration with large language model assistants (Claude, Anthropic). The AI contributed to experimental design, Python code implementation, data analysis, and manuscript drafting. All scientific decisions, theoretical motivations, and final interpretations were made by the human author, who takes full responsibility for the content.

## Code and Data Availability

All code, configuration files, and output data are available at: https://github.com/unicome37/poset_phase. Archived version: https://doi.org/10.5281/zenodo.18980657.

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